Evaluating the Effects of Place-Based Policy: Evidence from Opportunity Zones and the Low Income Housing Tax Credit

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Abstract

I study the impact of the Opportunity Zones Tax Credit, a federal place-based investment incentive passed as part of the 2017 tax reform, on the development of Low Income Housing Tax Credit subsidized housing projects using a difference-in-differences design. Across my main specifications, I find no statistically significant effect of tract-level Opportunity Zone designation on either LIHTC projects or the number of assisted units constructed in a tract. Additionally, my estimates suggest that the program did not differentially effect either of the two main sub-categories of LIHTC subsidies. My preferred point estimates are small and negative, and precise enough to rule out effect sizes larger than 0.1σ at the 95% confidence level. LIHTC projects were already disproportionately located in Opportunity Zone Census tracts before the policy intervention due to strong selection of projects by tract poverty levels, and my analysis suggests that the Opportunity Zones program did not shift additional subsidized housing projects into OZ designated tracts.

1 Introduction

There is growing interest in economics and policy circles in the use of place-based policies to address persistent geographic disparities in income, employment, poverty, wealth, and access to economic opportunity (Bartik 2020; Austin, Glaeser, and Summers 2018). This new policy focus on place stands in contrast to economists' traditional skepticism towards place-based programs as inefficient and distortive compared to programs that target disadvantaged individuals.

The Opportunity Zones tax credit, passed as part of the 2017 Tax Cuts and Jobs Act, is the latest in a series of federal policies designed to incentivize investment in disadvantaged communities. Compared to previous place-based programs, such as Enterprise Zones or the New Markets Tax Credit, Opportunity Zones are much more geographically extensive (covering 10.7% of Census tracts and 9.7% of the US population) and administratively simpler to take advantage of. The OZ tax credit provides several forms of capital gains tax relief to investors who invest unrealized capital gains into an asset or qualified business located in a Census tract designated as an Opportunity Zone.

Research on the Opportunity Zones program is thin, primarily due to the newness of the program. Existing studies generally fall into one of two categories: those that use IRS tax data to document the extent and type of investment in Opportunity Zones, and studies that perform program evaluations using a variety of outcome measures, including real estate development, employment, earnings, and house prices. I contribute to the literature by studying the impacts of OZ designation on Low Income Housing Tax Credit (LIHTC) project development. LIHTC is the federal government's primary tool for incentivizing the creation of subsidized housing units.

Importantly for this paper, LIHTC investors can take advantage of both programs by stacking the OZ incentive on top of LIHTC credits (Head 2020; Soroka and Milder 2019). Affordable

housing advocates and professional service firms that specialize in assisting LIHTC developers have highlighted the compatibility of the two incentive programs: "[Combining] LIHTCs with OZs is one of the most attractive uses of the OZ Incentive, as most designated OZs have troubled residential housing markets ... The programs blend well, particularly in light of the rigidness of the "substantial improvement" requirement under the OZ Incentive" (Williams 2022, 499).

There has been meaningful takeup of the OZ incentive: using de-identified IRS tax data, Coyne and Johnson (2022) find that through 2020, cumulative OZ investment flows have totaled $48bn, a majority of which is in the real estate sector. Previous work has found significant impacts of tract OZ designation on real estate development (Wheeler 2022). This provides evidence of a "first stage" effect of OZ designation on investment, particularly in real estate development.

I extend this existing research by looking specifically at OZ's impacts on affordable housing development. The program's effect on overall real estate development, combined with its compatibility with existing affordable housing subsidies, raises the possibility that LIHTC developers could use OZ funds as an alternative to conventional debt-based financing. Affordable housing projects are an example of the type of "social impact" development that the program's boosters hoped that the tax incentives would promote, and studying the program's impact on this specific type of real estate development gives us a better idea if the program is meeting its stated goals on this dimension (Wessel 2021). I also contribute to the broader literature on the impacts of the Opportunity Zones tax credit, which is still an open question as the impacts of the program continue to play out (Fikri and Glasner 2023).

Specific features of the LIHTC also mean that the results are of broader relevance to the literature on place-based development policy. In general, design-based program evaluations of place based development policies cannot distinguish between geographic substitution and the inducement of new investment (akin to the income and substitution effects of a price

change in classical microeconomic theory) without imposing further assumptions or model structure.[1] Disentangling the two effects is an important concern for the evaluation of place based policies, which have the goal of maximizing the inducement of new investment per subsidy dollar while minimizing distortions of the location of infra-marginal projects that would have gone forward without the subsidy.

The LIHTC program provides a natural laboratory to perform this type of decomposition due to the inelastic supply of certain LIHTC credits. The LIHTC "9 percent" credits are allocated to states by the IRS following statutory formulas, and the credits are in excess demand (Scally, Gold, and DuBois 2018). Thus the supply of these credits is inelastic, and any effect of OZ designation on LIHTC developments utilizing 9% credits can be interpreted as purely a geographic substitution effect. Comparing this effect to the OZ program's effect on overall LIHTC developments is informative about the extent to which OZ developments incentivize new economic activity versus shifting inframarginal projects to take advantage of the subsidy.

To evaluate the impact of the Opportunity Zones program on LIHTC project development, I compare Census tracts that were designated as Opportunity Zones to tracts that were eligible under program rules but not designated using a difference-in-differences design. Across my preferred specifications, I find no statistically significant effect of tract-level Opportunity Zone designation on either LIHTC projects or the number of assisted units constructed in a tract. My point estimates are small and negative, and precise enough to rule out effect sizes larger than 0.1 standard deviation units of the dependent variable.

In Section 2 I summarize the existing literature on the OZ program and detail how my work contributes to the literature. In Section 3, I review relevant background and policy details of the Opportunity Zones program as well as the Low Income Housing Tax Credit. In Section 4, I outline the data I employ in my analysis and present data patterns that motivate

[1] To complete the microeconomic theory analogy, we cannot decompose the income and substitution effects of a price change without assumptions on the shape of the decision maker's utility function.

my empirical analysis. In Section 5, I describe my empirical approach in detail and discuss identification and potential sources of bias. In Section 6, I present the results of my empirical analysis as well as additional extensions and robustness checks. Section 7 concludes.

2 Existing Literature

Due to the young age of the Opportunity Zones program (it was passed at the end of 2017 and implemented in 2018), there is a relatively thin literature about the program compared to existing place-based policies such as the Enterprise Zones program or the New Markets Tax Credit, and even fewer studies attempt to evaluate the policy impacts of the program. I provide a brief overview of the existing literature on Opportunity Zones, as well as federal place-based polices more broadly, and detail my contribution to our understanding of the program.

The literature on federal place-based policies generally found modest, if any, impacts on outcomes for disadvantaged communities. The two largest federal place-based programs over the past few decades have been the New Markets Tax Credit (NMTC) and the Empowerment Zones (EZ) program.

The NMTC is a competitively administered program run by the Treasury department. Over the course of 2001-2019, almost $60 billion of NMTC credits were approved; a wide variety of types of projects can qualify for the credits; and credits are frequently won by Community Development Financial Institutions or other mission lenders (Qian 2019). Theodos et al. (2022) study the impacts of the program, finding that the credits increased population and jobs, but that inflows of new residents with college degrees were sufficiently large to account for the increases in population and employment. These results suggest that the program's benefits may not have accrued to existing residents of areas that benefited from the tax credits.

The EZ program designated a small number of distressed areas as eligible for employment tax credits and block grants. The tax credits subsidized the employment of zone residents inside the designated EZ, and the block grants were intended to be used for investment in social services. Busso, Gregory, and Kline (2013) study the effects of the first round of the Empowerment Zones program, finding that the program successfully increased employment and wages for zone residents, without inducing population growth or increasing housing rents.

I break down the literature on the OZ program into three categories: papers that describe the OZ selection process and the characteristics of OZ designated tracts, papers that use IRS tax data to document the type and extent of investment into OZs, and papers that evaluate the impact of OZs on a number of different economic and sociodemographic outcomes.

Immediately following the rollout of the OZ program, researchers and policy analysts sought to understand the characteristics of the tracts that were designated as opportunity zones and investigate whether states had maximized the potential of their selections (Theodos, Meixell, and Hedman 2018). They found that designated tracts were negatively selected on a number of sociodemographic indicators compared to eligible but non-designated tracts, and also found that designated tracts were potentially marginally positively selected on access to credit, as measured by real estate and small business lending. Frank, Hoopes, and Lester (2022) investigated the political economy of the states' OZ assignment process, finding that tracts whose local representatives were of the same party as the states' governor had a higher probability of being designated as Opportunity Zones.

Another set of papers has used IRS tax data to document the nature, extent, and spatial distribution of Opportunity Zone investments, as well as the characteristics of the taxpayers investing through the program (Kennedy and Wheeler 2022; Coyne and Johnson 2022). Using the complete set of electronic and paper tax returns, Coyne and Johnson (2022) report that Opportunity Zones had received a cumulative $48 billion dollars in investment through the program through the end of 2020. Although these are considerable sums of money, they

are well short of the trillion dollar figures that the program's creators were projecting at the time of the bill's passage. Coyne and Johnson (2022) also show that the majority (over 60%) of OZ funds have been invested in real estate projects. Kennedy and Wheeler (2022) use the set of electronically filed returns to examine patterns of OZ investment. They find that investment has been highly concentrated: over 60% of OZ tracts have not received any investments. Moreover, they show that investment has been concentrated in higher-SES OZ tracts.

Finally, there is a growing number of papers conducting program evaluations of the Opportunity Zones tax credit. Arefeva et al. (2021) study the effect of OZ designation on employment using an establishment-level panel, finding a 3-4% increase in employment at the establishment level. Wheeler (2022) uses building permit records from a set of large cities and finds that there was a large increase (2.9 percentage points, or 20%) in new construction in OZ tracts compared to eligible but non-designated tracts in the same cities. Freedman, Khanna, and Neumark (2021) use restricted ACS microdata to study the effects of the program on zone residents. After accounting for pre-existing trends, they find no impact of OZ designation on zone resident's employment, earnings or poverty status. Chen, Glaeser, and Wessel (2023) study the effect of OZ designation on home prices and find a zero to small (0.5%) impact on home prices in OZ tracts.

The discrepancies in findings across the program evaluation papers is not driven by differences in research design: all use some form of difference-in-differences strategy to estimate the effects of OZ designation on different economic outcomes. Instead, the differences appear to be driven by the type of outcomes considered in the different studies; papers looking at broader socioeconomic outcomes or indirect effects such as home prices have tended to find null outcomes, while the studies that look at direct outcomes of the OZ investments (jobs and building permits) have found positive effects of the program. My reading of the literature as it stands is that the OZ program has successfully attracted some investment to

disadvantaged Census tracts, but that the program has yet to generate benefits that have spillovers to existing residents.

I contribute to the literature evaluating the impacts of the Opportunity Zones tax credit by considering a particular type of real estate development: Low Income Housing Tax Credit developments. Studying LIHTC developments allows me to assess whether the OZ program is meeting its stated goals regarding "social impact" investments. Additionally, the inelastic supply of the 9% LIHTC credit allows me to decompose the overall effect of LIHTC development into geographic substitution and new supply effects, which is of broader interest to the literature on evaluating the impacts of place-based economic development policies.

3 Policy Background

In this section, I lay out the history and relevant policy details of each of the two programs I study: Opportunity Zones and the Low Income Housing Tax Credit.

3.1 Opportunity Zones (OZ)

Opportunity Zones (OZs) is a place-based tax incentive program passed as a small component of the 2017 Tax Cuts and Jobs Act (TCJA). The goal of the program is to increase investment in historically underinvested communities across the United States. The OZ program works by giving a variety of tax incentives to investors (either corporations or individuals) who invest *unrealized capital gains* in projects in Census tracts that were designated under the program as Opportunity Zones. I discuss the OZ designation process in detail below.

The program provides three distinct tax incentives for investors who invest existing unrealized capital gains in a project located in a designated OZ through a vehicle registered with the IRS as a Qualified Opportunity Fund (QOF).

1. Deferral of capital gains taxes on the invested capital gains until 2026.
2. Step up in basis on the original investment that generated the capital gains:
    - If the asset is held for at least 5 years, the original basis is stepped up by 10%
    - If the asset is held for at least 7 years, the original basis is stepped up by 15%
3. No capital gains taxes on the new investment if it is held for at least 10 years

All three tax incentives serve to reduce investors' capital gains tax liability, increasing the net-of-tax return for investing in Opportunity Zones, which is the mechanism by which the policy attempts to attract investment into tracts designated as OZs. The subsidy provided by the three tax benefits is meaningful. Coyne and Johnson (2022) argue that the largest component of the subsidy is the exemption of the investment from additional capital gains taxes if it is held for 10 years, although this figure is sensitive to the assumed rate of return on the investment. In an illustrative example, they find that the exemption of new gains from additional taxation accounted for about 75% of the total tax benefit, which itself amounted to $250k on a hypothetical $1m Opportunity Zone investment.

There are very few restrictions placed on investments to qualify for the OZ tax credit. Investments may be made in tangible assets, such as real estate, or businesses that conduct "substantially all" of their business in an Opportunity Zone. The only restriction on the type of assets is the exclusion of "sin businesses" such as liquor stores, gambling facilities, and golf courses or country clubs. Additionally, unlike prior place-based policies such as the NMTC, there are only very minor reporting requirements. QOFs self-certify semi-annually by filing a form with the IRS reporting total assets held and the amount of those assets that are held in Opportunity Zone property.

Funds can be invested in a wide variety of different sectors, including real estate, housing, infrastructure, and existing or start up businesses located within an OZ. In practice, about 60% of OZ funds have been invested in real estate projects, while the next largest sector is finance and insurance at about 20% of invested funds (Coyne and Johnson 2022).

3.1.1 Opportunity Zone Selection Process

The process for designating Census tracts as OZs happened very quickly after the passage of the TCJA at the end of 2017, and allowed for wide discretion on the part of state governors, who were empowered to select tracts as OZs from a large set of eligible tracts: in total, 56% of Census tracts in the US were eligible to be designated as OZs. The TCJA defined Census tracts as eligible to be OZs if they are "low income communities" under either of two criteria:

1. Poverty rate $\geq 20\%$
2. Median family income $\leq 80\%$ of max(state med. income, metro area med. income)

Additionally, tracts contiguous to low income tracts are eligible if their median family income does not exceed 125% of the eligible tract's income. In practice, very few tracts (less than 2%) that were designated as OZs qualified solely through contiguity to a low-income tract.

The governor of each state was allowed to designate 25% (or 25 tracts, if the number of eligible tracts in their states was fewer than 100) of the eligible tracts as Opportunity Zones. Governors were given wide scope to select the designated tracts from the eligible set, and the decisions were required to be made with very quick turnaround from the passage of the law (only about 120 days).

Tracts that were designated as OZs on average had higher poverty rates, lower median household income, lower education rates, and lower employment rates than eligible tracts that were not selected as OZs (Theodos, Meixell, and Hedman 2018; Frank, Hoopes, and Lester 2022).

This pattern is largely in keeping with the goal of the program to direct investment dollars to disadvantaged communities. However, some analysts raised concerns that the designated tracts were only "minimally targeted" towards census tracts with low levels of preexisting investment flows (as measured by commercial, real estate, mortgage, and small business lending), suggesting that the tracts selected already were not especially disadvantaged in

terms of pre-existing access to capital (Wessel 2021; Theodos, Meixell, and Hedman 2018).

3.2 Low Income Housing Tax Credit (LIHTC)

The Low Income Housing Tax Credit (LIHTC) is the federal government's main policy tool to encourage the development of low-income rental housing. LIHTC was passed as a part of the 1986 Tax Reform Act, and since then over 3.5 million assisted units have been placed into service across more than 52,000 different housing projects (Public and Affordable Housing Research Corporation and National Low Income Housing Coalition 2023).

LIHTC works by offering developers generous income tax credits in return for meeting unit affordability requirements. The subsidized units must typically be affordable to residents making 60% of the area median income, and units must be kept affordable for at least 30 years, after which they may be converted to market rate.[2] Although projects are not required to be 100% affordable, states' credit allocation guidelines typically incentivize this; 68% of LIHTC projects have 100% affordable units.

Over time, a specialized ecosystem of developers that focus on LIHTC projects has emerged. Developers raise capital from investors by selling the credits they are allocated to investors in return for an equity investment. Developers frequently supplement the capital raised by selling the tax credits with debt, either through a traditional mortgage loan or other financing from government sources (Scally, Gold, and DuBois 2018).

LIHTC has two primary forms of tax credits. 9% Credits are primarily used for new project construction and are allocated to states annually by the IRS on a formula basis, and allocated to developers by state housing authorities via a competitive application process. 4% Credits are available for projects using primarily tax-exempt bond financing, and can be claimed without the competitive allocation process. In practice, 9% credits are typically used for

[2]Units are considered affordable to renters of a given income level if the rent does not exceed 30% of their income

new construction projects, while 4% credits are used to redevelop existing properties into affordable housing projects.

The tax credits constitute a generous subsidy to equity investors. Over the 10-year horizon that investors claim the credit, the 4% credit delivers roughly a 30% NPV subsidy, and the 9% credit delivers a roughly 70% NPV subsidy. Additionally, projects located in particularly disadvantaged tracts designated as Qualified Census Tracts get an additional 30% increase in the value of the subsidy, which increases the subsidy provided by the 9% credit to a 91% NPV subsidy.

Of particular relevance to this project, equity investors in LIHTC projects can also claim the OZ tax credit if the investment is made through a Qualified Opportunity Fund (QOF) (Soroka and Milder 2019).

3.2.1 LIHTC Development Timeline

The LIHTC development cycle starts with the annual allocation of tax credits to state housing finance agencies (HFAs) by the IRS. Each state HFA then solicits proposals from developers, which are scored according to criteria set by the state HFA. HFAs typically have one or two application cycles each year (Keightley 2023). After the HFAs receive the applications, they decide which proposals will receive the credits; this process varies state-by-state according to their credit allocation rules. At this point, the state reports the winning projects to the Department of Housing and Urban Development, which are then recorded in the HUD LIHTC database with the project's *credit allocation date*.

After being allocated the credits, the developer needs to secure financing in order to begin construction on the project. A large portion (55% on average) of financing is covered by the sale of tax credits to investors; the developers then secure their remaining financing through a combination of traditional mortgage debt and other public or private financing sources (Scally, Gold, and DuBois 2018). Once construction has been completed, the project

is certified and the state HFA then transfers the credits to the developer. At this point, the project's *put in service* date is recorded in the HUD LIHTC database. The timing of the financing and construction stages varies across projects. However, developers are generally allowed two years to complete the project upon credit allocation (Keightley 2023); this places a relatively tight ceiling on the time between credit allocation and when the project is put in service.

4 Data

I use three primary data sources in this project: a database of Census tract-level Opportunity Zone designations, Census tract demographics from the American Community Survey, and data on LIHTC subsidized projects from the Department of Housing and Urban Development (HUD).

4.1 Opportunity Zone Status

For my main measure of treatment, whether or not a Census tract was designated as an Opportunity Zone, I use a geospatial database from HUD, which records for every Census tract:

1. Whether or not the tract was eligible to be designated as an Opportunity Zone under the criteria described in Section 3.1.1
2. If the tract was eligible to be designated:
    1. The reason for the tract's eligibility
    2. Whether or not the tract was designated as an Opportunity Zone

I use this database to construct my main analysis sample and define my treatment indicator. Table 1 tabulates the number of tracts in each of the following mutually exclusive categories:

ineligible, eligible but not designated, and designated Opportunity Zone. Throughout my analysis, I restrict the sample to the 50 states and DC.[3] In total, of the 72,848 Census tracts in the United States, 41,201 (56%) were eligible to be designated as Opportunity Zones, and 7,809 (11% of the total) were actually designated as OZs. OZ Census tracts account for 10.8% of the US population.

Table 1: Counts of Cenus Tracts by OZ Treatment Status

Tract Category	N	Percent
Ineligible	31,647	0.43
Not Designated	33,392	0.46
Designated	7,809	0.11

4.2 Census Tract Characteristics

I link the HUD Opportunity Zones database with tract-level demographics from the 2013-2017 5-year American Community Survey estimates. This is the same data that were made available to state policymakers when they were making Opportunity Zone designations in the spring and early summer of 2018.

Designated tracts are negatively selected on observable socioeconomic characteristics, including tract poverty rates, education levels, and median household income. Figure 1 plots the distributions of several tract characteristics across designated OZ and eligible but non-designated tracts, illustrating this negative selection on observables.

In addition to the demographic data from the ACS, I also include measures of other tract characteristics. I include a measure of tract urbanicity from the USDA Economic Research Service's Rural-Urban Commuting Area Codes (RUCA). I invert the scale of the measure so that a value of 1 denotes a rural tract, and a value of 10 denotes a tract in a metropolitan area

[3]All tracts in Puerto Rico were designated as Opportunity Zones and the territories were not eligible for the program.

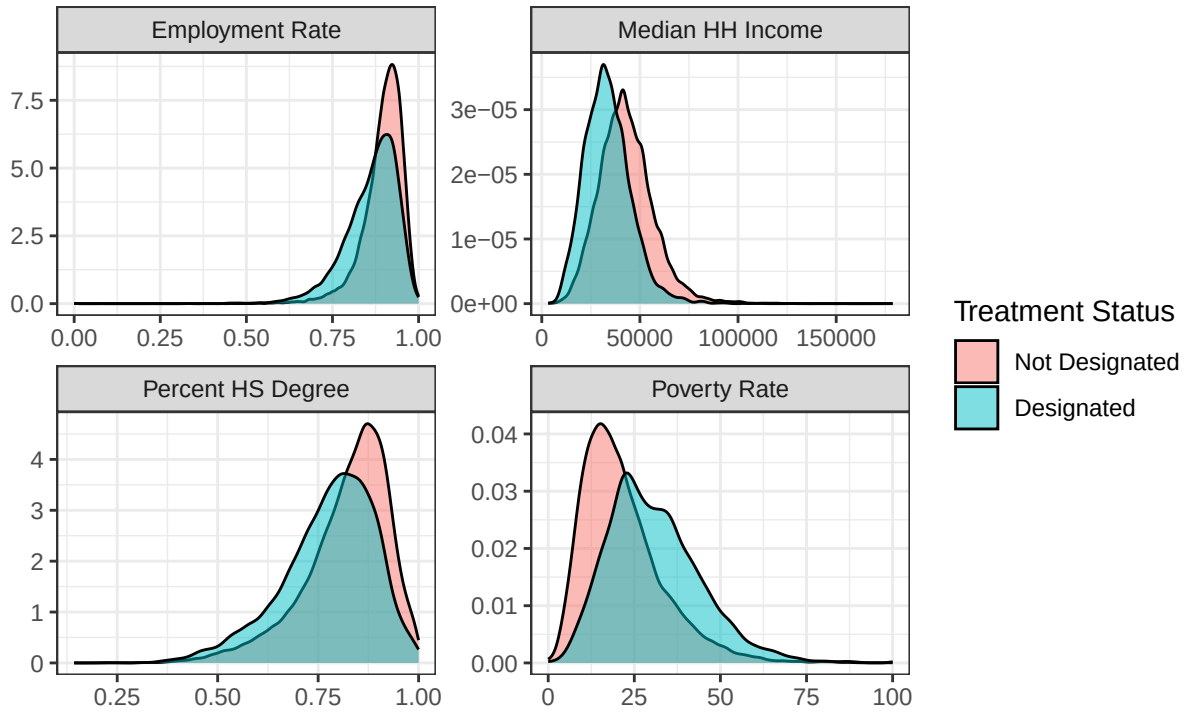


Figure 1: Demographics of Designated vs. Eligible but Non-designated Tracts

core. I also include measures of pre-treatment tract-level investment and sociodemographic change from Theodos, Meixell, and Hedman (2018).

4.3 Low Income Housing Tax Credit

For my outcome measures, I use data on counts of new LIHTC subsidized housing projects and assisted units. I source the LIHTC data from the National Housing Preservation Database (NHPD), a database covering federally-assisted affordable housing projects created by the Public and Affordable Housing Research Corporation and the National Low Income Housing Coalition. The NHPD sources its LIHTC data from HUD's publicly-available LIHTC project database, but performs additional data cleaning, de-duplication, and validation of the project geolocation data. In addition, the NHPD provides records at the subsidy–rather than property–level, so it is possible to observe multiple rounds of subsidies over time or the

combination of multiple federal housing subsidy programs at the same project.

In addition to my main outcome measures (counts of projects and assisted units), the database also records the type of LIHTC tax credit (4% or 9%) the project received, the type of developer (for profit or not-for-profit), whether or not the project is 100% affordable or mixed affordable and market rate units. I use this additional information in for followup analyses to explore the mechanisms driving my main results.

For my analysis, I take the LIHTC data, which are geolocated to the street address level, and match projects to 2010 Census tracts, which are the unit of treatment assignment for the Opportunity Zones program. I then aggregate across Census tracts and years to create a tract-year level panel that I use for my main analyses. Following Gold et al. (2018), who also use the NHPD data on LIHTC subsidized projects, I start my panel in 2000 (rather than 1987, the first year of the LIHTC program) due to data missingness in the records from the earliest years of the LIHTC program. I construct my panel using the credit allocation date, which measures when projects are approved by their state's Housing Finance Authority, rather than the placed-in-service date, which measures when the project construction has been completed. This choice helps alleviate the issue of construction lags delaying the appearance of LIHTC projects in the data.

Although my analysis sample has over 25 thousand LIHTC subsidized projects, subsidies are a relatively rare event at the Census tract-year level. LIHTC projects are concentrated across space in lower income areas, and in most years a given tract does not receive a new LIHTC project. Across my entire panel, 1.7% of tract-year observations have a LIHTC project, while 22.1% of tracts receive any LIHTC projects across the entire sample period. Restricting to the set of Census tracts that were eligible to be designated as Opportunity Zones, 2.4% of tract-year observations have a LIHTC project and 27.8% of tracts see any projects over the entire sample period.

All results in the paper are substantively similar when run using the HUD LIHTC database

instead of the NHPD database.

4.4 Data Patterns

Figure 2 plots a binscatter of total LIHTC projects, aggregated to the Census tract level across the entire sample period against tract poverty rates. LIHTC projects are strongly selected on tract poverty rates. Additionally, there is a jump in projects at the 25% poverty threshold for Qualified Census Tracts, where the LIHTC subsidy increases discontinuously. This replicates a well-known result in the LIHTC literature (Baum-Snow and Marion 2009). This strong sorting of LIHTC projects into high-poverty tracts is driven both by federal regulations and states' allocation guidelines for the competitively-administered 9% tax credits (Scally, Gold, and DuBois 2018).

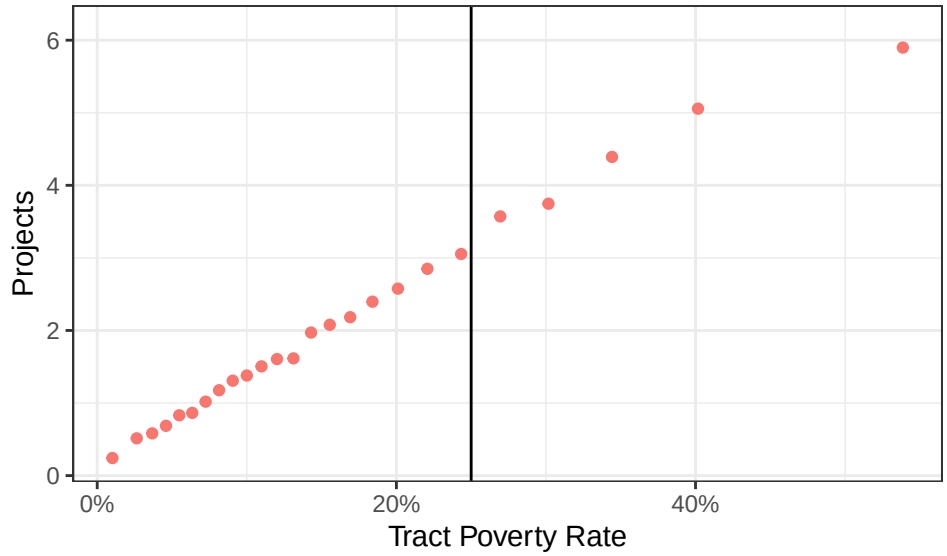


Figure 2: Total LIHTC Projects vs. Tract Poverty Rate

Table 2 reports the results from a tract-level regression of total LIHTC projects on tract poverty rate and QCT status, which confirm the visual pattern in the binscatter plot above. A tract being designated as a QCT is associated with 1.4 extra LIHTC projects over the sample period, and increasing the poverty rate by 10 percentage points is associated with

0.78 extra projects. The strong selection of LIHTC projects into higher-poverty tracts, as well as the strong additional impact of QCT status on LIHTC projects, are both important to account for in the main empirical analysis.

Table 2: Tract-Level Poverty Regression

	LIHTC Projects
Poverty Rate	7.895
	(0.158)
QCT	1.407
	(0.051)
N	72,814

This selection pattern of LIHTC projects into higher-poverty tracts interacts with the pattern that higher-poverty tracts were also more likely to be designated as Opportunity Zones to generate large differences in levels of LIHTC projects by OZ tract status: designated, eligible but not designated, and ineligible. Figure 3 plots the average number of LIHTC projects per tract, grouped by tract OZ status, across the sample period.

Consistently across the sample period, tracts designated as Opportunity Zones receive about twice as many LIHTC projects as eligible but not designated tracts, and five times as many as ineligible Census tracts. This stratification of LIHTC projects by tract category is driven by the positive relationship that both LIHTC projects and OZ designated tracts have with tract-level poverty rates.

Due to reporting lags in the HUD LIHTC database, I restrict my panel to end in 2020. The data in the LIHTC database nominally extend through 2022, but the number of projects in the database in 2021 and 2022 drops to near-zero. This is driven by reporting lags documented by Gold et al. (2018); *not* an actual decrease in the number of credits apportioned to states by the IRS or allocated by the state Housing Finance Authority. I discuss the importance of truncating the sample to exclude the periods with the reporting lags below in Section 5.3.

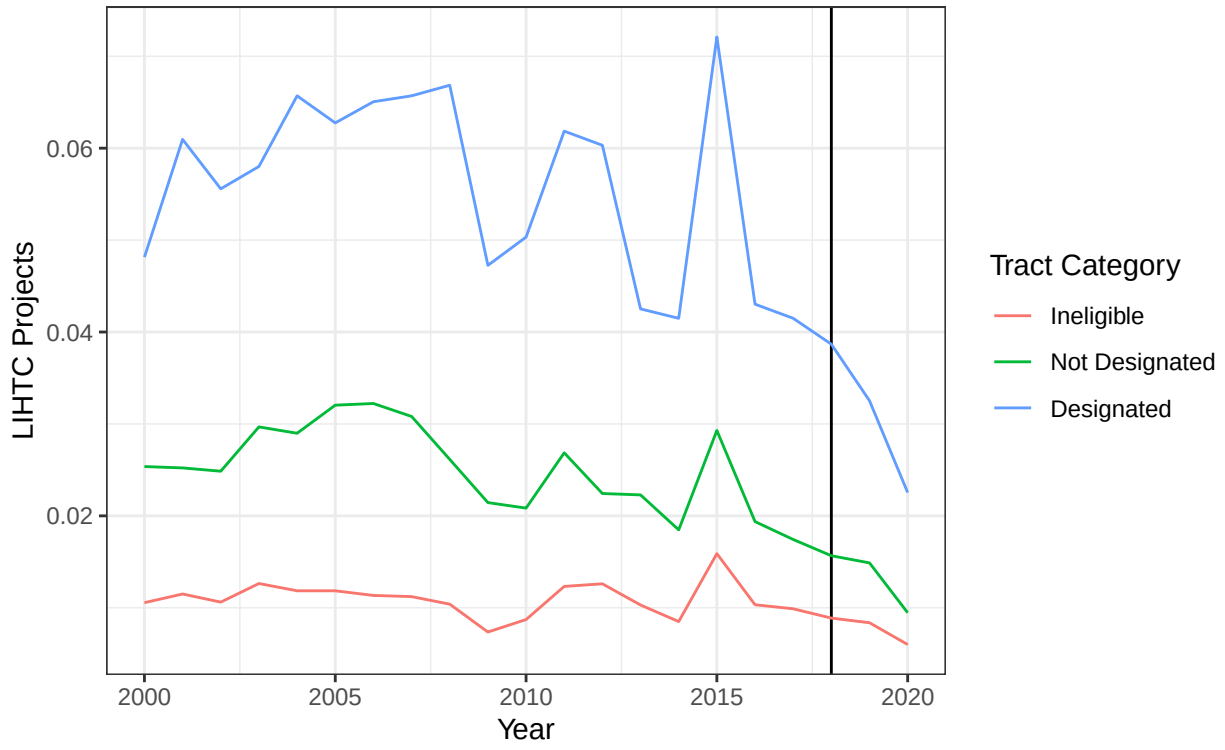


Figure 3: LIHTC Projects by Tract Category

5 Empirical Strategy and Identification

I employ a difference-in-differences framework to estimate the effect of Opportunity Zone designation on LIHTC development. The treatment group in my setting is Census tracts that were selected by states' governors to be designated as Opportunity Zones. For the untreated or control group, I use the set of tracts that were eligible to be designated as Opportunity Zones but not designated. This follows other studies in the literature, such as Chen, Glaeser, and Wessel (2023) and Freedman, Khanna, and Neumark (2021), who also employ difference-in-difference designs to evaluate the OZ program. Ineligible tracts are very different in terms of both socioeconomic characteristics as well as levels of LIHTC development, and therefore are a less credible control group than the eligible but non-designated tracts.

Due to the federal nature of the OZ program, all units in my policy context are treated at the same time. Additionally, it is reasonable to assume that there were no anticipation effects

whereby LIHTC developers shift development activity into OZ tracts, knowing that the tracts will later be treated. The tax credit was a low-profile component of the much larger TCJA, and there was little policy discussion or media coverage of the provision prior to the passage of the bill in December 2017. State governors were given only a few months in the spring of 2018 to designate tracts as Opportunity Zones, and the program went into effect in July 2018 (Wessel 2021). These institutional features of my policy context mean that I can work with a simplified modeling framework, ignoring issues such as staggered rollout and treatment anticipation effects that have been discussed in the new literature on difference-in-differences models.

The notation below follows Callaway and Sant'Anna (2021), suppressing the group indices because I only have a single treatment group. $Y_t(j)$ for $j \in \{0, 1\}$ denotes the realization of outcome variable Y in time period t, with $j = 1$ denoting (potentially counterfactual) treated outcomes and $j = 0$ denoting (potentially counterfactual) untreated outcomes. $D \in \{0, 1\}$ represents treatment status, and X is a vector of control variables.

Under appropriate identifying assumptions, the difference-in-differences design recovers a (time-specific) average treatment effect on the treated (ATT):

$$ATT(t) = E[Y_t(1) - Y_t(0)|D = 1, X] \tag{1}$$

The ATT is the difference in outcomes (here LIHTC development) *for units that were selected into the treatment*, compared to a counterfactual scenario in which those same units were not given the OZ treatment. The key identifying assumption that delivers the ATT interpretation of the difference-in-differences results is a conditional parallel trends assumption:

$$E[Y_t(0) - Y_{t-1}(0)|D = 1, X] = E[Y_t(0) - Y_{t-1}(0)|D = 0, X] \ \forall t \tag{2}$$

This conditional parallel trends assumption in Equation 2 states that counterfactual untreated outcomes for the treatment group ($D = 1$) would have evolved identically (in levels) to the observed outcomes for the untreated group ($D = 0$) across all time periods, both pre- and post-treatment, conditional on some set of variables X.

Conditioning on tract characteristics is critical in this context, because as I showed above, both tract poverty rates and QCT status are positively correlated with both LIHTC development and OZ status. This correlation would potentially violate an unconditional parallel trends assumption. For that reason, it is important to include both tract poverty rates and QCT status in X, the vector of conditioning variables for the conditional parallel trends assumption. Additional variables in the conditioning set include tract sociodemographic characteristics, urbanicity, measures of pre-treatment investment and socioeconomic change, as well as state fixed effects. I describe the variables in the conditioning set and the propensity score model in greater detail in Section A.1.

5.1 Estimation

The standard estimation approach for a difference-in-differences design is the two-way fixed effects (TWFE) model:

$$Y_{it} = \alpha_i + \delta_t + \beta_{TWFE} D_{it} + \varepsilon_{it} \tag{3}$$

Here, α_i are the unit (Census tract) fixed effects and δ_t are time period fixed effects. D_{it} is a treatment indicator, and is equal to 1 for designated OZ tracts after 2018 (and 0 otherwise). Under a parallel trends assumption akin to the one in Equation 2, but not conditioning on the vector of tract characteristics X, β_{TWFE} is an unbiased estimator of the ATT. However, as I discuss above, the strong joint selection of both LIHTC development and OZ designation on tract poverty rates undercuts the credibility of the unconditional parallel trends assumption

in my setting.

In addition to the TWFE estimates, I also use the doubly-robust (DR) estimator provided by Callaway and Sant'Anna (2021) for computing period-specific $ATT(t)$s and aggregate the effects to an overall ATT.[4] Following Sant'Anna and Zhao (2020), the DR model requires specifying both a propensity score model for units' treatment status, and a model for the evolution of the outcomes of the comparison group. The DR estimator is consistent if *either* the propensity score or outcome model are correctly specified. I discuss the specification of the propensity score model and show that there is meaningful overlap in the estimated propensity scores in Section A.1. I also report aggregate treatment effects, which in the case with a single treatment group corresponds to a simple average of the period-specific $ATT(t)$.

The primary advantage of using the Callaway-Sant'Anna estimator in a setting without staggered treatment rollout is that it allows me to relax the unconditional parallel trends assumption required for the TWFE estimates to be consistent and condition on pre-treatment covariates that predict both treatment and LIHTC development outcomes, which I believe makes the identifying assumption much more plausible in my context given the strong joint selection of treatment and the outcome variable on covariates such as tract poverty rates and QCT status. Additionally, the modeling of the propensity score allows the estimator to make "better" comparisons by putting more weight on control units that are more similar to the treated units.

5.2 Standard Errors

I cluster the standard errors at the state level. Abadie et al. (2023) argue that in design-based frameworks where treatment assignment differs between clusters, it its appropriate to cluster the standard errors at the level of treatment assignment. For the Opportunity Zones program,

[4] I use the `did` R package, written by the authors to implement the estimators presented in the paper, to estimate the DR model

treatment assignment (OZ designation) was done at the state level by the governor or a delegated agency, so in my results I report standard errors that are clustered at the state level. Callaway and Sant'Anna (2021) provide a multiplier bootstrap estimator for the standard errors of the DR model that allows for cluster dependence in the model residuals.

5.3 Reporting Lags and Parallel Trends Violations

The HUD LIHTC Database, which provides the source data for the NHPD, suffers from a reporting lag of 3-4 years. This results in zero or near-zero counts of LIHTC projects and units in the last few years of the dataset.

Due to the difference in levels between the treated and non-treated tracts at baseline, a decrease to zero in both groups at the end of the sample period creates a mechanical violation of the parallel tends assumption that underlies my differences-in-differences strategy. Consider the following simplified example to illustrate the issue:

The blue lines represent the outcomes for the treated group, and red lines outcomes for the untreated group. The true post-treatment outcomes are represented by the dashed lines, and the true ATT in this example is zero. Suppose however, that the "observed" data feature reporting lags, where the outcomes both fall to zero in the post-period. The level decline in the outcome variable is larger for the treated units, which have higher levels of the outcome variable at baseline. The ATT computed using the "observed" data with false zeros due to reporting lags is -2, which is biased downward from the true ATT. The direction of the bias introduced by data lags will in general depend on the difference in levels between the treated and untreated units before the reporting lags kick in.

In my case, the level of LIHTC developments at baseline is substantially higher in OZ tracts than in eligible but not designated tracts, so including time periods affected by reporting lags will bias my estimates of the treatment effects of the OZ program downwards, in the

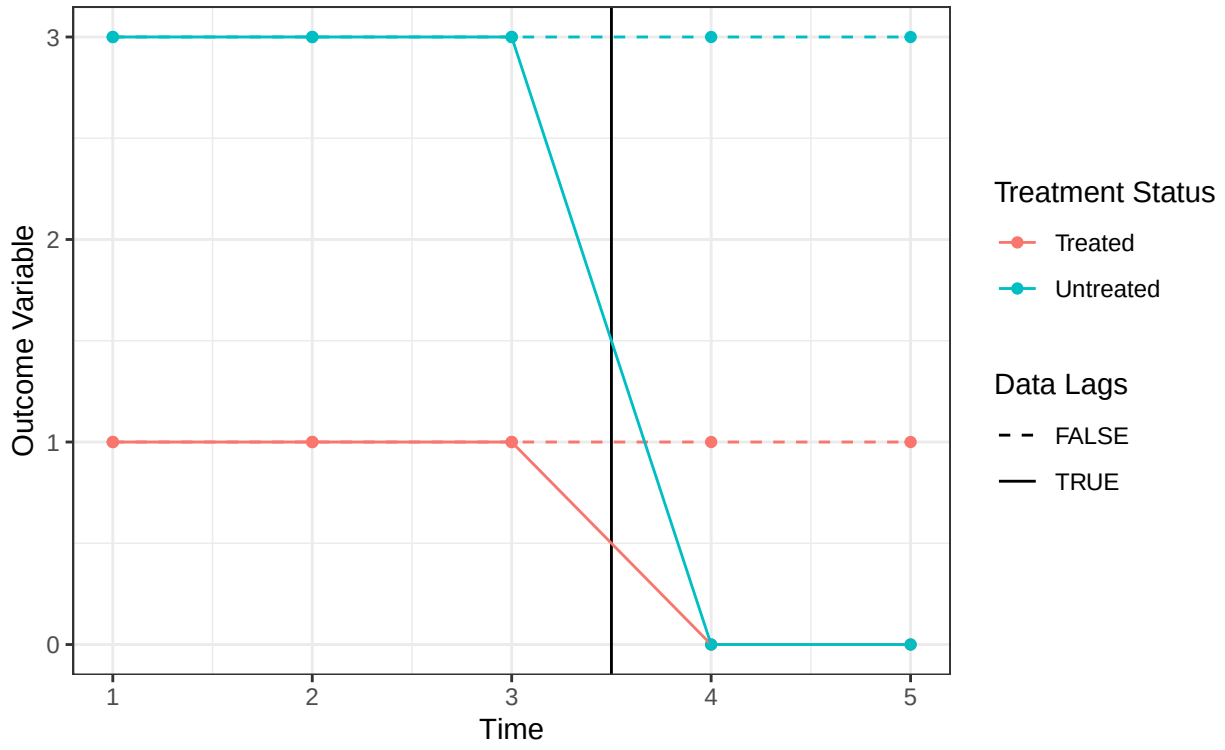


Figure 4: Reporting Lags Mechanically Violate Parallel Trends Assumption

same way as in the example above. As a consequence, I drop the final 3 years in the dataset that are affected by the LIHTC reporting lags.

6 Results

Table 3 presents the estimated ATTs (and associated standard errors) for my main difference-in-difference models. The six columns of the table are broken down across two outcomes (LIHTC projects and the number of subsidized units created by those projects) and three subsets of LIHTC credits: all credits, the 4% credits, and the 9% credits. The top panel of the table presents the estimates from the TWFE model, and the bottom panel the estimates of the aggregated ATT from the Callaway-Sant'Anna doubly-robust model. All standard errors in the table are clustered at the state level.

For the models in columns (1) and (4) that estimate the effect of OZ designation on LIHTC

Table 3: Difference-in-Difference Estimates

	Dependent variable:					
	Projects			Num. Units		
	All	4%	9%	All	4%	9%
	(1)	(2)	(3)	(4)	(5)	(6)
A. TWFE						
Treat	-0.013	0.000	-0.006	-0.650	0.063	-0.412
	(0.002)	(0.001)	(0.001)	(0.162)	(0.092)	(0.077)
B. Callaway-Sant'Anna						
ATT	-0.005	-0.002	-0.003	-0.468	-0.297	-0.173
	(0.005)	(0.003)	(0.003)	(0.401)	(0.329)	(0.206)
N	40,840	40,840	40,840	40,840	40,840	40,840
Mean of DV	0.029	0.008	0.012	2.085	0.884	0.628

[a] Model standard errors clustered at the state level.

development using the entire sample of LIHTC credits, the point estimates of *ATT* are consistently negative across the two dependent variables (counts of LIHTC projects and subsidized units created) and across the two different models. The point estimates for the TWFE model are each larger and more precisely estimated than the corresponding estimates from the Callaway-Sant'Anna DR model. 95% confidence intervals for the TWFE estimates exclude zero, while the confidence intervals for the estimated *ATT*s from the DR model include zero. My preferred specification is the Callaway-Sant'Anna model, because it explicitly allows for conditioning on pre-treatment characteristics, notably the tract poverty rate, that predict both the probability of treatment and subsequent outcomes.

Turning to the models that estimate effects of the program separately for the 4% and 9% tax credits, it does not appear that the overall effects are driven by either type of tax credit, or that the (absence of) an overall effect is masking competing effects of the two types of credit that go in opposite direction. Focusing on the Callaway-Sant'Anna estimates, for each of the outcome variables the estimates for the individual credit types are each roughly half of the overall effect size and not statistically distinguishable from each other. This suggests that

each of the two credits contributes equally to the overall (non) effect of OZ designation on LIHTC development outcomes.

In the context of my setting where the overall outcome is the sum of two component outcomes, the estimated treatment effect on the overall outcome should be the sum of the estimated treatment effects on the two component outcomes. The intuition is different to the case of performing analyses on a set of mutually exclusive subgroups (such as analyzing the effects of a labor market training program separately on men and women), in which case the overall treatment will be a weighted average of the subgroup treatment effects. To see this, consider a simple difference-in-differences setting. Let $Y_{it}^O = Y_{it}^1 + Y_{it}^2$, and consider the following specification for the outcomes:

$$Y_{it}^1 = \alpha_i^1 + \delta_t^1 + \tau^1 D_{it} + \varepsilon_{it}^1$$
$$Y_{it}^2 = \alpha_i^2 + \delta_t^2 + \tau^2 D_{it} + \varepsilon_{it}^2$$

Summing the two outcomes, it is clear that the treatment effect on the overall outcome will be the sum of the treatment effects on the two sub-outcomes:

$$Y_{it}^O = (\alpha_i^1 + \alpha_i^2) + (\delta_t^1 + \delta_t^2) + (\tau^1 + \tau^2) D_{it} + (\varepsilon_{it}^1 + \varepsilon_{it}^2)$$

The estimates in 3 are generally consistent with the intuition from this simple setup, especially for the Callaway-Sant'Anna estimates. The estimated effects on total LIHTC developments is roughly the sum of the estimated effects on 4% and 9% credits.

One possible explanation for the apparent lack of impacts of the OZ program on LIHTC developments is that the program stimulated *applications* for LIHTC credits, but that these additional applications were not approved. It is impossible to test this hypothesis directly because the state HFAs and the HUD database only report successful applications that are allocated the LIHTC credits, not the entire set of applications. However, for this story to be

consistent with my estimates from 3, it would have to be the case that the OZ designation stimulated new applications for the 9% credits but not the 4% credits, which are not subject to the same state application process. In addition, it would require that the state HFAs disproportionally accepted non-OZ application for the 9% credits to maintain the same level of approved projects in OZ and non-OZ tracts. This strikes me as extremely implausible, so I conclude that it is unlikely that the absence of effects on LIHTC developments is driven solely by changes in credit application behavior by developers.

Figure 5 shows an event-study style plot of the six models from the lower panel of Table 2. The post-period ATT estimates are in blue, while the pre-period estimates are red. These event-study style plots provide some visual evidence to support the conditional parallel trends assumption that underlies the causal interpretation of the ATT estimates from the DR model. While the assumption is fundamentally untestable because it is about counterfactual outcomes for the treated group that are never observed in the post-period, the plots are consistent with the implications of the assumption in the pre-treatment periods: there are no significant difference in outcomes (conditional on the underlying propensity score model) between the treated and untreated tracts.

6.1 Effect Sizes

I focus my discussion of effect sizes on the estimates from the Callaway-Sant'Anna doubly robust model, which is my preferred specification. A common measure of standardized effect sizes is the ratio of the estimated effect (here my aggregated ATT estimates) to the standard deviation of the dependent variable in the sample. The estimated effect sizes are small compared to the variance present in the raw data: the standard deviation of the number of LIHTC developments per tract-year in the analysis sample is 0.196, and the standard deviation of the number of LIHTC assisted units per tract-year is 19.75. The standardized effect sizes are on the order of 0.01-0.07 standard deviation units, and are estimated with

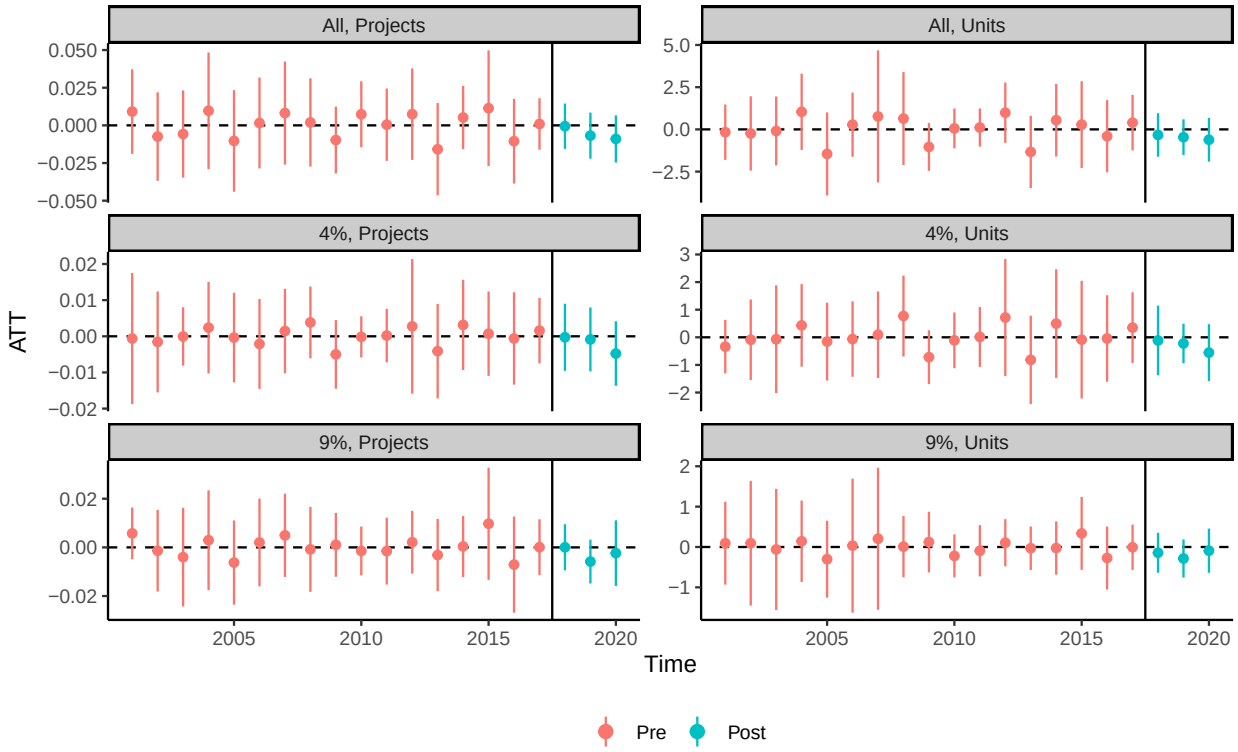


Figure 5: Callaway-Sant'Anna Time-Specific Treatment Effects

sufficient precision to rule out effect sizes greater than 0.1 standard deviations of the dependent variable using a 95% confidence interval.

6.2 Extensions and Robustness

In this section I estimate variations of my primary model on different subsets of the data to explore mechanisms that may be driving the lack of impact of the OZ program on LIHTC developments.

6.2.1 Tract Socioeconomic Status

I first investigate whether the OZ program has differential impacts based on tract SES. Kennedy and Wheeler (2022) find that 60% of OZ tracts did not receive any investment through the end of 2020, and that the tracts that did receive investments were higher

SES than tracts that had not received investments. Without access to the IRS data, I cannot directly observe tract-level investment. Instead, I create an SES index as the first principal component of the variables they find are correlated with investment: population, unemployment, household income, racial composition and educational achievement. Table 4 reports the principal component loadings for the variables in the SES model.

Table 4: SES Variable Loadings

Variable	Loading
Population	0.09
Median HH Income	0.51
Poverty Rate	-0.55
% White	0.44
% HS	0.48

After creating the SES index, I then split my sample into tracts with above and below median SES index levels and re-estimated the models in column 1 of Table 3 on the high and low SES samples. Table 5 reports the estimates from these models.

Table 5: Treatment Heterogeneity by Tract SES

	High SES	Low SES
A. TWFE		
Treat	-0.012	-0.007
	(0.003)	(0.003)
B. CS DiD		
ATT	-0.010	-0.004
	(0.011)	(0.019)
N	20,442	20,398
Mean of DV	0.019	0.039

[a] Model standard errors clustered at the state level.

The estimates are on the same order as those from the main models in Table 3: estimated effects are small and negative, and not distinguishable from each other or zero. If anything, the larger negative effects are for the higher SES tracts compared to the lower SES tracts.

This suggests that the overall lack of an effect of OZ designation on LIHTC development is not masking a positive effect for a smaller subset of tracts that are more likely to have received OZ investment.

6.2.2 Pre-Existing LIHTC Investment

I also consider if there is heterogeneity by whether tracts had seen LIHTC developments before the introduction of the OZ policy. This sort of "intensive margin" effect could arise if LIHTC developers have experience in particular geographic locations and find it less costly to propose developments in areas that they are already familiar with. To test this channel, I estimated the models from column 1 of Table 3 on the subset of tracts that pre-treatment LIHTC developments.[5] Table 6 reports estimates of the TWFE and Callaway-Sant'Anna models on this subsample of the data:

Table 6: Extensive Margin Estimates

	TWFE	C-S
Treat	-0.006	
	(0.003)	
ATT		-0.009
		(0.009)
N		11,030
Mean of DV		0.104

[a] Model standard errors clustered at the state level.

Again, the estimates are small and negative, and not statistically distinguishable from the main estimates in Table 3. This exercise suggests that the overall lack of effect I find is not masking an impact on a smaller subset of tracts that we might expect ex-ante to have a higher likelihood of attracting LIHTC investments.

[5] 27% percent of tracts had pre-treatment LIHTC developments

7 Conclusion

I find relatively precisely-estimated null effects of Census tract Opportunity Zone designation on LIHTC development activity. These null effects are consistent across outcomes (counts of LIHTC developments and the number of LIHTC subsidized units) as well as the different types of LIHTC credits. Additionally, I fail to find heterogeneous impacts of the program by tract SES or the presence of pre-treatment LIHTC developments in the tract. My results suggest that the OZ program has not been effective in promoting affordable housing construction through the LIHTC program, despite its overall impact on real estate development.

There are a combination of institutional features of the two programs that interact to explain the apparent lack of takeup of the OZ tax credit to finance LIHTC developments. Although some features of the two programs are aligned (notably 10 year holding requirements for claiming the LIHTC credits matching the 10 year holding requirement to forgo capital gains taxes on the OZ investment), there are conflicting features from the perspective of both potential Opportunity Zone investors as well as LIHTC developers. LIHTC development is administratively complex, and is typically undertaken by a developer that specializes in subsidized housing and supported by an extensive team of accountants and tax lawyers (Scally, Gold, and DuBois 2018). For an investor considering taking advantage of the OZ credit, the closest substitute is market rate rental housing developments, which have comprised a large share of total OZ investment to date, and are substantially less administratively complicated than LIHTC developments. From the perspective of the LIHTC developers, their modal equity investor is a bank seeking to satisfy Community Reinvestment Act requirements, and the business model of banks in the United States means that they typically do not have substantial sums of unrealized capital gains required to take advantage of the OZ credit. This means that LIHTC developers' typical investor pool is not interested in making investments OZ-eligible.

The absence of an overall effect of OZ designation on LIHTC development, and the 9% credits

in particular, prevents me from performing a decomposition exercise that exploits the fact that the LIHTC 9% credits are allocated inelastically via a statutory appropriations rule. This was an attractive feature of the setting, because it would have permitted the decomposition of an aggregate effect on LIHTC development into geographic substitution and new supply channels, a question that is important in the evaluation of place-based development policies but typically not answerable using standard program evaluation techniques.

Despite the surface compatibility and apparent appeal of the Opportunity Zones tax credit as a means to boost net of tax returns on LIHTC investments, which already frequently occur in OZ-eligible Census tracts due to the alignment of program rules on tract poverty rates, deeper technical issues appear to have limited take up of the OZ credit for funding LIHTC developments. To the extent that policymakers wish to encourage development in disadvantaged and historically under-invested communities, it is important to consider in detail how potential new programs would interact with existing incentives to ensure they are compatible.

A Appendix

A.1 Propensity Score Model

An important component of the Callaway-Sant'Anna doubly-robust DID estimates is the underlying propensity score model of tract treatment. The model requires estimating a propensity score using pre-treatment characteristics, and the estimated propensity scores are then used to weight observations when computing the overall ATT. Sant'Anna and Zhao (2020) show that this class of doubly-robust estimator is consistent if either the propensity score model or the outcome regression model are correctly specified. In this appendix, I describe the specification of the first-stage propensity score model, present the model estimates, and demonstrate that there is substantial overlap in the estimated propensity scores for treated and non-treated units.

To specify my propensity score model, I follow Frank, Hoopes, and Lester (2022), who build a model of Opportunity Zone designation to study the political economy of the selection process. The key variables that they include are baseline tract demographics from the ACS and characteristics of the states' selection process, such as whether the selection was carried out by the governor's office or delegated to an independent economic development agency. Rather than including the suite of variables that characterize the selection process, I flexibly control for these cross-state differences by including state fixed effects in the regression model. In addition to demographic characteristics from the ACS, their model also controls tract urbanicity, and the socioeconomic change and investment scores from Theodos, Meixell, and Hedman (2018). In addition to the variables used by Frank, Hoopes, and Lester (2022), I also add LIHTC QCT status to the model, because as I showed in Section 4, QCT status is correlated with both LIHTC development outcomes and tract treatment status.

Equation 4 gives the regression equation for the model described above:

$$D_i = Population_i + MedianHHIncome_i + PovertyRate_i + EmploymentRate_i + PctWhite_i + PctEducated_i + Urban_i + InvestmentScore_i + SocioeconomicChange_i + QCT_i + State_i + \varepsilon_i \tag{4}$$

Here D_i represents tract treatment status, with $D_i = 1$ for tracts that are designated as OZs, and $State_i$ are state fixed effects. I estimate the model as a logistic regression, becase the estimated propensity scores are required to be in the $(0, 1)$ interval to estimate the ATTs. Table 7 presents estimates of the propensity . Column 1 reports the coefficient estimates on the logistic scale, while column 2 gives the average marginal effects, with standard errors calculated via the delta method. For comparability with Table 4 in Frank, Hoopes, and Lester (2022), I report the estimates of a linear probability using the same specification in Column 3.

Figure 6 plots the distribution of estimated propensity scores for the treated and untreated tracts. The mode of the distribution of scores for treated tracts is higher than the mode of the untreated distribution. Additionally, the treated distribution has a much fatter right tail than the untreated distribution. Importantly, although the right tail is thinner for the untreated distribution, it is still extends out quite far, and there is common support over almost the entire distribution of estimated propensity scores.

The unconditional mean of the estimated propensity scores is 19%, which by construction matches the unconditional mean treatment probability in the data. The mean estimated treatment probability for the treated tracts is 28.4%, while the mean for untreated tracts is 16.7%.

Table 7: Propensity Score Model Estimates

	Logit	Average Marginal Effect	LPM
Population	0.046	0.006	0.005
	(0.008)	(0.001)	(0.001)
Median Hh Income	-0.044	-0.006	-0.004
	(0.002)	(0.001)	(0.000)
Poverty Rate	1.540	0.208	0.428
	(0.191)	(0.047)	(0.027)
Employment Rate	-3.209	-0.433	-0.602
	(0.263)	(0.037)	(0.039)
Pct White	-0.005	-0.001	0.009
	(0.066)	(0.009)	(0.009)
Pct Educated	-0.447	-0.060	-0.052
	(0.157)	(0.020)	(0.023)
Urban	-0.078	-0.011	-0.010
	(0.006)	(0.002)	(0.001)
Dec Score	0.056	0.008	0.008
	(0.005)	(0.001)	(0.001)
SE Flag	0.655	0.102	0.086
	(0.083)	(0.018)	(0.012)
QCT	0.259	0.036	0.048
	(0.040)	(0.008)	(0.006)
N	40,840	40,840	40,840
Mean of DV	0.19		0.19

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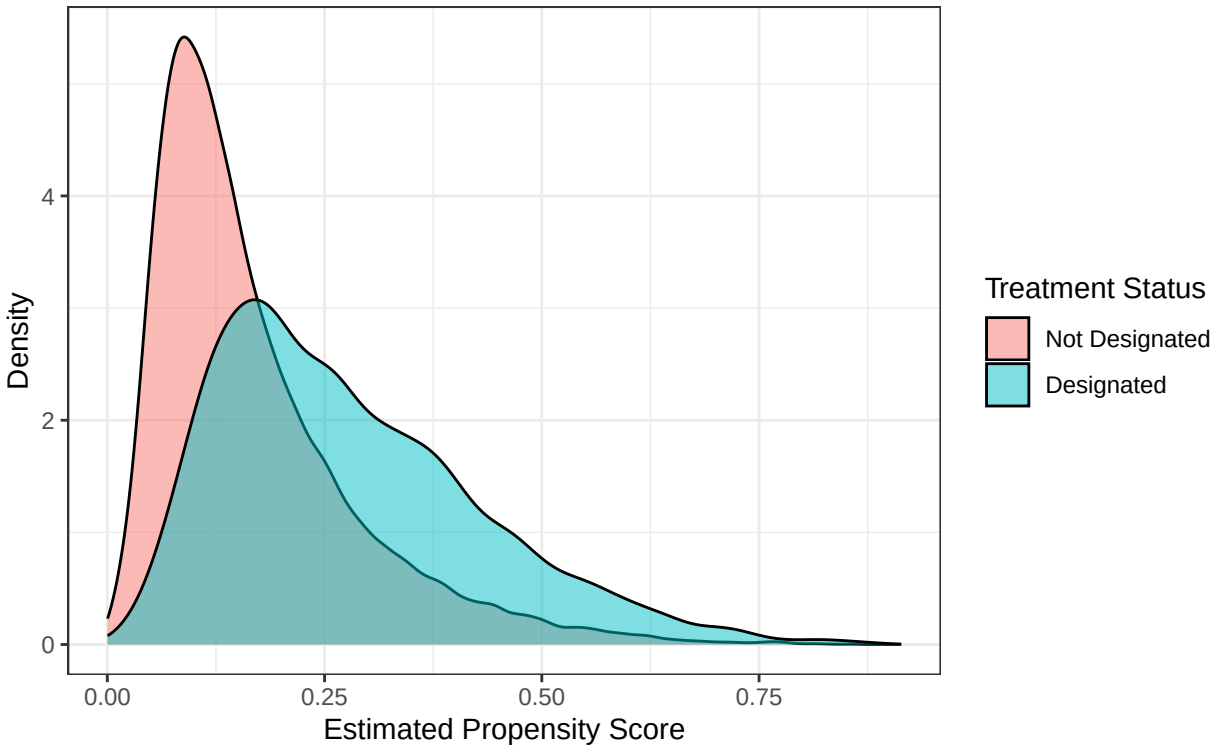


Figure 6: Distribution of Estimated Propensity Scores

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